

# StayEase Boutique Hotels

## Booking Cancellation Prediction

*What is the business about?*

StayEase Boutique Hotels is a hospitality company that provides short-stay accommodation for leisure, business, family, conference, and international guests. The company earns revenue through room bookings, meal plans, parking fees, and hotel upgrades.

*What problem is the business trying to solve?*

The main business challenge is booking cancellation. When a guest cancels close to the arrival date, the hotel may lose revenue because the room cannot always be resold. At the same time, if the hotel assumes too many guests will cancel, it may overbook rooms and create poor customer experiences.

*What decision can machine learning help the business make?*

Machine learning can help the business decide to predict whether a booking is likely to be cancelled. This allows managers to take reasonable actions, such as sending reminder emails, offering flexible rebooking options, adjusting overbooking policies, or prioritizing waitlisted customers.

*What is the target variable in the dataset?*

The target variable in the dataset is `Booking_Cancelled`, where 1 means the booking was cancelled and 0 means the booking was not cancelled.

*What are the input features?*

The input features include booking behavior variables such as `Lead_Time_Days`, `Booking_Channel`, and `Deposit_Type`; customer history variables such as `Repeated_Guest`, `Previous_Cancellations`, and `Loyalty_Points`; trip details such as `Weekend_Nights`, `Weekday_Nights`, `Adults`, `Children`, and `Room_Type`; and revenue-related information such as `Average_Daily_Rate`, `Meal_Plan`, and `Parking_Requested`.

*Why is this prediction useful for the business?*

This prediction is useful for the business because it helps identify high-risk bookings, take appropriate actions, review procedures and policies, improve business strategies, and support revenue management.

## Business Interpretation of Model Results

### How well did the Decision Tree model perform?

The Decision Tree model showed moderate performance in predicting booking cancellations. It achieved approximately 57.89% accuracy, 42.86% precision, 55.56% recall, and 48.39% F1-score. These results indicate that the model was able to identify some cancellation patterns, but its overall predictive performance is still limited. The confusion matrix also showed several incorrect predictions, meaning the model still struggled to consistently distinguish between cancelled and non-cancelled bookings.

### Is there a large difference between training accuracy and testing accuracy?

No, there is not a very large difference between the training accuracy and testing accuracy. The training accuracy is approximately 76%, while the testing accuracy is around 58%. Although the training accuracy is higher than the testing accuracy, the gap is still moderate rather than extreme.

### Does the model show signs of overfitting? Why or why not?

The model shows some signs of overfitting because the training accuracy is higher than the testing accuracy. The controlled Decision Tree achieved about 76% training accuracy but only around 58% testing accuracy. This means the model performed better on the data it learned from compared to new booking data.

To further assess overfitting, the results were compared with an overfitted Decision Tree model. The overfitted model achieved 100% training accuracy but only about 51% testing accuracy. This large gap shows that the model memorized the training data too closely and could not generalize well to unseen booking data. Compared to the overfitted model, the controlled Decision Tree showed better balance between training and testing performance, indicating that limiting the tree depth helped reduce severe overfitting.

### Which evaluation metric is most important for this business problem?

For this hotel booking cancellation problem, recall is the most important evaluation metric because the hotel wants to identify as many cancelled bookings as possible. Recall measures how many actual cancelled bookings were correctly detected by the model.

This is important because missing cancelled bookings may lead to empty rooms, lost revenue, and poor room planning. A higher recall helps the hotel detect more possible cancellations early so managers can take action, such as contacting guests or adjusting room availability.

However, precision is also important because low precision means the model may incorrectly predict that many bookings will be cancelled even when the guests actually plan to arrive. This can lead to unnecessary business actions, such as sending extra reminders or making incorrect overbooking decisions.

### What do false positives mean in this business context?

A false positive means the model predicts that a booking will be cancelled, but the guest actually arrives at the hotel. In this situation, the hotel may incorrectly assume that the room will become available and make unnecessary business actions, such as sending additional reminder messages, adjusting room inventory, or making inaccurate overbooking decisions. Too many false positives may reduce operational efficiency and affect customer experience because the hotel is reacting to cancellations that never actually happen.

### **What do false negatives mean in this business context?**

A false negative means the model predicts that a booking will not be cancelled, but the guest actually cancels the reservation. In this situation, the hotel may assume the guest will arrive and fail to prepare for the cancellation in advance. This can lead to empty rooms, missed opportunities to resell the booking, lost revenue, and less effective room and staffing management. Too many false negatives can reduce the hotel's ability to respond early to possible cancellations.

### **Which features were most important in the Decision Tree model?**

Some of the most important features in the Decision Tree model were Average\_Daily\_Rate, Lead\_Time\_Days, Loyalty\_Points, Booking\_Channel, Repeated\_Guest, Room\_Type, Deposit\_Type, and Weekday\_Nights. These features had higher importance scores, meaning they played a bigger role in helping the model predict whether a booking was likely to be cancelled or not.

### **How can these important features help the business make better decisions?**

These important features can help StayEase Boutique Hotels better understand which bookings are more likely to be cancelled and why. For example, long lead times, certain booking channels, or specific deposit types may indicate a higher cancellation risk, while repeated guests and customers with loyalty points may be more likely to complete their bookings.

By using these insights, hotel managers can make better business decisions, such as improving room planning, adjusting overbooking strategies, sending reminder messages to high-risk customers, and prioritizing reliable guests. This can help reduce revenue loss, improve room occupancy management, and support better customer service.

### **What is one possible limitation or bias in the model or dataset?**

One possible limitation is that the dataset is simulated and may not fully represent real-world hotel booking behavior. Important real-life factors such as weather conditions, travel disruptions, local events, emergencies, or sudden policy changes are not included in the dataset. In addition, the model may learn historical business patterns that could unintentionally create bias toward certain customer groups or booking channels.

### **Why should human judgment still be used when making business decisions based on model results?**

Human judgment should still be used because the Decision Tree model showed only moderate performance and still produced several incorrect predictions. The model may incorrectly predict that a booking will be cancelled when the guest actually arrives, or fail to identify actual cancellations. Because of these prediction errors, relying only on the model could lead to poor business decisions, such as inaccurate overbooking, unnecessary customer actions, or missed revenue opportunities.

Hotel managers can use their experience and business knowledge to review the model's predictions and consider other real-world factors that are not included in the dataset, such as weather conditions, special events, customer situations, or travel disruptions. Therefore, the model should be used as a decision-support tool rather than the only basis for making business decisions.